# A Multilingual NLP Analysis of Movie Reviews

*Tokenization, Embeddings, Attention and Parameter-Efficient Fine-Tuning across Korean, Chinese and English*

Mid-term Project Report


## Abstract

This report documents a controlled comparative study of natural-language processing across three typologically distant languages — Korean, Chinese, and English — using movie-review sentiment as a common downstream task. Three public corpora (NSMC, Douban, IMDB) were sampled to yield two thousand reviews per language with a balanced positive-negative split, giving six thousand documents in total. Using a shared multilingual backbone (DistilBERT-base-multilingual-cased), we examined four stages of a modern NLP pipeline: subword tokenization, static and contextual embeddings, layer-wise attention, and parameter-efficient fine-tuning. The goal was not to chase leaderboard numbers but to ask, at each stage, why a particular result is what it is, and what that tells us about the interaction between language structure and the model. The study confirms a familiar but important pattern: tokenization dominates the behavior of CJK pipelines, contextual embeddings capture polysemy that static vectors cannot, attention does not automatically focus on contrastive conjunctions, and roughly one percent of the parameters — placed correctly — recover most of the gains of full fine-tuning.


## 1. Introduction

Multilingual natural-language understanding is an unusually good microscope for studying what a pretrained model actually knows. When the same architecture is asked to classify sentiment in three scripts, with three different levels of morphological inflection and three different writing conventions, the places where the model stumbles reveal the places where language-specific structure and learned representations fail to meet. The project described here uses movie reviews as a common genre. Reviews are short, opinionated, syntactically varied, and rich in contrastive constructions such as "good acting but weak plot", which makes them a useful testbed for attention analysis.

Three parallel sentiment corpora were chosen to cover the Korean, Chinese, and English writing traditions. Korean movie reviews (NSMC) are typically short and heavily inflected; Chinese Douban reviews vary in length and rely on explicit sentiment vocabulary; English IMDB reviews are often long and use syntactic connectives to build polarity. Fairness across languages is a central concern of the study, so dataset sizes, models, and training schedules were held as close to identical as possible. Differences that remained in the output can then be attributed, with reasonable confidence, to linguistic structure rather than to experimental artefacts.

The remainder of the report follows the logical order of a typical NLP pipeline. Section 2 describes how the data was assembled and cleaned. Section 3 compares baseline tokenizers with a shared BPE tokenizer. Section 4 contrasts static Word2Vec with contextual BERT embeddings. Section 5 visualizes attention

matrices in two representative sentences per language. Section 6 reports sentiment-classification accuracy for a frozen baseline, a fully supervised fine-tune (SFT), and a LoRA adapter. Section 7 examines the errors made by the best model; Section 8 draws the cross-lingual comparison together; Sections 9 and 10 close with conclusions and known limitations.

## 2. Data

Three public corpora were used. NSMC (Naver Sentiment Movie Corpus) provides Korean reviews with binary labels derived from star ratings. Douban Movie Comments provides Chinese reviews with a one-to-five star scale that was collapsed to binary by treating four or five stars as positive and one or two as negative; three-star neutral comments were dropped. The Stanford IMDB corpus supplies English reviews that are already labelled positive or negative. All three datasets were loaded from the local copies delivered with the assignment.

Text cleaning followed a uniform protocol. HTML tags, URLs, and repeated whitespace were removed; empty strings and duplicates were dropped; documents shorter than five characters were discarded; and the English text was lowercased to match common IMDB preprocessing practice. Non-Hangul Korean rows were filtered out to avoid lines that slipped into the corpus in other scripts. Labels were unified to the two-class set {positive, negative} across all languages.

The project brief explicitly requires multi-movie coverage to prevent topic bias from confounding linguistic analysis. For Chinese, the source data carried a movie-name field, so a stratified sample was drawn that preserves coverage across 28 distinct titles — each movie contributes roughly the same number of positive and negative comments. For Korean and English, the source datasets did not expose reliable movie-level metadata, so the sample is drawn uniformly at random from the full pool; because NSMC and IMDB are each already pooled from thousands of films, the resulting mixture is still broad rather than single-title. After sampling, every language contributes exactly one thousand positive and one thousand negative documents, giving a perfectly balanced six-thousand-document corpus. Figure 1 summarizes the distribution.

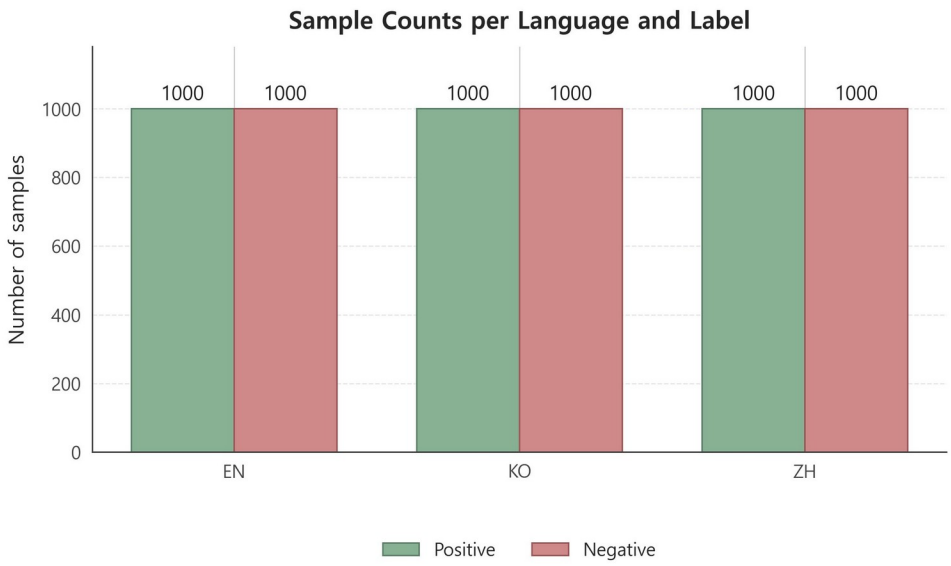


**Figure 1. Sample counts per language and label (balanced at 1,000/1,000 after stratified sampling).**

**Table 1. Dataset composition after cleaning and stratified sampling.**

| Language | Source | Positive | Negative | Total | Multi-movie |
|---|---|---|---|---|---|
| Korean | NSMC | 1,000 | 1,000 | 2,000 | pool-level |
| Chinese | Douban | 1,000 | 1,000 | 2,000 | 28 titles |
| English | IMDB | 1,000 | 1,000 | 2,000 | pool-level |
| Total | — | 3,000 | 3,000 | 6,000 | — |

The data was finally shuffled with a fixed random seed (42) and serialized to data.csv with the four required fields: text, label, source, language (an auxiliary movie column is also kept). The same seed is reused for every downstream split so that tokenizer statistics, embedding experiments, and classifier training are all conducted on identical inputs.

## 3. Tokenization

Tokenization is the first place where the three languages diverge sharply. English uses spaces to mark word boundaries, so a whitespace split already produces reasonable tokens. Chinese is written without spaces and relies on a word segmenter (Jieba) to recover lexical units. Korean has spaces but is heavily agglutinative, with suffixes glued to stems; a morphological analyser (Kiwipiepy) is the linguistically appropriate tool. The experiment compares each of these language-specific baselines against a single shared subword tokenizer — the multilingual BPE vocabulary that ships with distilbert-base-multilingual-cased — so that we can see what is gained and what is lost when one model is forced to serve all three scripts.

Four statistics were computed for every language-tokenizer pair: the average number of tokens per document, the average character length of a token, the compression ratio (tokens divided by characters in the raw text), and the out-of-vocabulary rate. The OOV rate for each baseline was estimated by holding out 20% of each language's texts, building the vocabulary on the remaining 80%, and counting tokens at test time that did not appear in the training vocabulary; for BPE the OOV rate is the fraction of [UNK] tokens, which is structurally near zero because BPE can always fall back to single characters. All four metrics are plotted side-by-side in Figure 2, and Table 2 lists the numerical values.

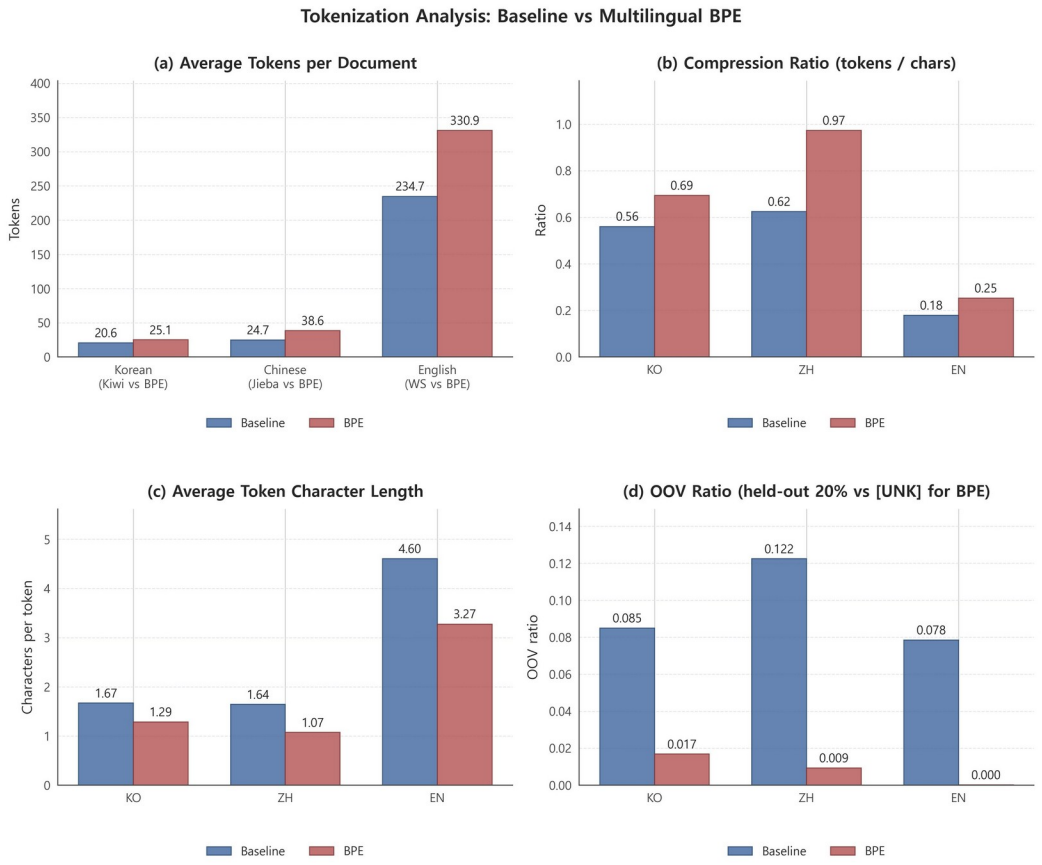


**Figure 2. Tokenization comparison: language-specific baselines vs multilingual BPE across four metrics.**

**Table 2. Tokenization statistics by language and method (same documents, two tokenizers).**

| Language | Baseline | Base tok. | BPE tok. | Base char/tok | BPE char/tok | Base OOV | BPE OOV |
|---|---|---|---|---|---|---|---|
| Korean | Kiwi | 20.6 | 25.1 | 1.67 | 1.29 | 0.085 | 0.017 |
| Chinese | Jieba | 24.7 | 38.6 | 1.64 | 1.07 | 0.122 | 0.009 |
| English | Whitespace | 234.7 | 330.9 | 4.60 | 3.27 | 0.078 | 0.000 |

The English numbers dominate the chart because IMDB reviews are long: even whitespace produces 234 tokens on average, and BPE pushes that figure to 330. The more informative dimension, however, is the

OOV rate. Whitespace tokenization leaves around 8% of English test tokens unseen because IMDB writers introduce a long tail of rare spellings, punctuation-joined words, and typos; BPE reduces this to essentially zero by splitting any unknown word into known subwords. The same pattern is visible in Korean and Chinese: Kiwi has an OOV of 8.5% and Jieba a striking 12.2%, while BPE cuts both to below 2%. This is the single most important property of subword tokenization — it resolves the open-vocabulary problem at the cost of a longer token sequence.

The compression ratios quantify that trade-off. For Chinese, the BPE compression ratio is 0.97, meaning the tokenizer produces nearly one token per character; Chinese characters are already semantic atoms and multilingual BPE treats most of them as individual tokens. The BPE subword ratio for Chinese is 0.3%, which confirms that almost no characters are joined into multi-character subwords under this tokenizer. For Korean the subword ratio is 57%, reflecting the fact that Hangul syllable blocks are frequently decomposed by the BPE merges. English sits in between at 14%. The practical implication is that the downstream transformer sees the same DistilBERT vocabulary across all three languages, but the amount of structure recoverable from a single token varies dramatically: a BPE token in Chinese is roughly a character (1.07 chars on average); in Korean it is a syllable or morph (1.29); in English it is a word stem or subword (3.27).

Which tokenizer is more appropriate depends on the task. For an interpretable linguistic analysis — error inspection, morpheme-level study — Kiwi and Jieba return recognizable words. For a downstream neural model with a fixed multilingual vocabulary, BPE's robustness to unseen strings outweighs its longer sequences, and it is the only choice compatible with multilingual BERT. The rest of this report therefore uses BPE for embeddings, attention analysis, and classification, while continuing to use Kiwi and Jieba when inspecting individual words.

## 4. Embedding

An embedding is the bridge between discrete text and continuous computation. The central question of this section is whether two different embedding philosophies — static (one vector per word type) and contextual (one vector per word token, depending on its surroundings) — produce recognizably different geometries. Word2Vec was trained from scratch on the cleaned corpora, with 100-dimensional vectors, a window of five tokens, fifteen epochs, and a minimum count of two. BERT embeddings were extracted from distilbert-base-multilingual-cased.

The nearest-neighbour results of Word2Vec already give a hint that training on two thousand documents is too little data to learn clean sentiment geometry. For Korean the neighbours of 재미 (fun) are 필요 (need), 남 (other), and 잊 (forget) — all with cosine 0.99. For Chinese, 精彩 (wonderful) lists 厉害 (awesome), 跟 (and), and 那个 (that one). Only English produces mostly sensible neighbours: great is close to wonderful (0.79), nice (0.73), brilliant (0.73), and terrible to horrible (0.85), awful (0.84), weak (0.82). The reason Korean and Chinese vectors collapse to near-uniform similarity is the small corpus — with only two thousand short documents per language there are simply not enough co-occurrences to separate the

semantic cone. English reviews are an order of magnitude longer, giving Word2Vec many more training pairs.

BERT embeddings do not suffer from this corpus-size problem because the model was pretrained on Wikipedia-scale data. Figure 3(a) projects 360 mean-pooled BERT sentence embeddings (120 per language) onto the first two principal components. Language identity is recovered cleanly with three separated clusters; the multilingual model has learned to encode language as a global feature before it encodes meaning. Figure 3(b) restricts attention to the English subset and colours the points by sentiment: positive and negative reviews are not separable in PC1-PC2 space. Two principal components capture variance that is dominated by language and topic, not by polarity. Sentiment signal exists in the embedding space, but it sits in directions that appear only in higher components — or, more precisely, in directions that a linear classifier on top of pooled CLS embeddings can pick up, as we will see in Section 6.

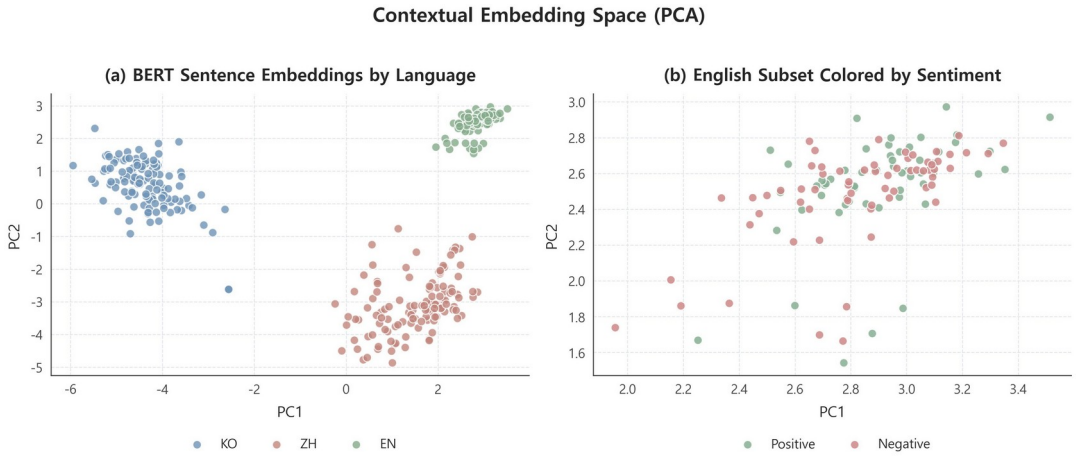


**Figure 3. PCA projection of BERT sentence embeddings: clear language clusters (left) but no sentiment separation within English (right).**

Figure 4 isolates a specific property: whether the embedding for the same word changes when the surrounding sentence changes. For three sentences per language, the target word (love in English, 好 in Chinese, 좋 in Korean) was embedded in context and pair-wise cosine similarities were computed. Word2Vec, by construction, produces the same vector every time the string appears — every pair therefore has similarity 1.00 (the left bars in the figure). BERT, by contrast, adjusts the contextual embedding according to the sentence: cosine similarity drops to 0.70–0.80 for English, 0.80–0.81 for Chinese, and 0.62–0.79 for Korean. The Korean result is particularly striking — 좋 appears in three genuinely different morphological environments (좋아요 / 좋지만 / 좋은) and BERT assigns them noticeably different vectors. This is the clearest illustration of why contextual embeddings matter for sentiment: the word 좋 is positive in 좋아요, but hedged in 좋지만 where it is followed by an adversative ending. A static embedding cannot represent that distinction at all.

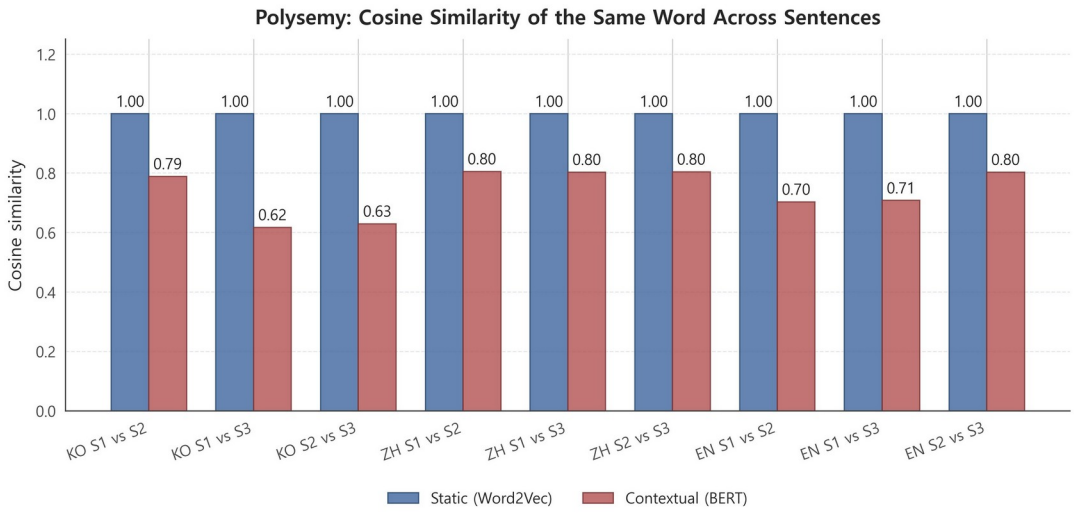


**Figure 4. Polysemy test: cosine similarity of the same target word across three different sentences (Word2Vec vs BERT).**

# 5. Attention Visualization

Attention is often described as the mechanism by which a transformer "decides what to look at". The project brief asks whether attention actually concentrates on the parts of a sentence that a human reader would consider decisive — in particular, whether the model attends more to a contrastive conjunction (but / 但是 / 지만) when one is present, compared to a neutral sentence without one. To test this, two sentences per language were constructed: one with an adversative clause, one without, matched for length. Attention at Layer 0 and Layer 5 (the last layer of the six-layer DistilBERT) was averaged across heads and rendered as a heatmap.

The English pair is shown in Figure 5. At Layer 0 the matrix is dominated by the [CLS] column: every query token attends heavily to [CLS] (values of 0.21–0.48), a well-known property of early transformer layers where [CLS] acts as a global information sink. At Layer 5 the pattern is very different: diagonal and local attention dominates, with tokens attending to themselves or to adjacent tokens. The contrastive word but receives 0.22 attention from the token the following it in the with-but sentence at Layer 5, but the sentiment verb enjoyed receives 0.42 attention from its following token this in the no-but sentence. In other words, in English the content word enjoyed is actually a stronger attention magnet than the functional word but.

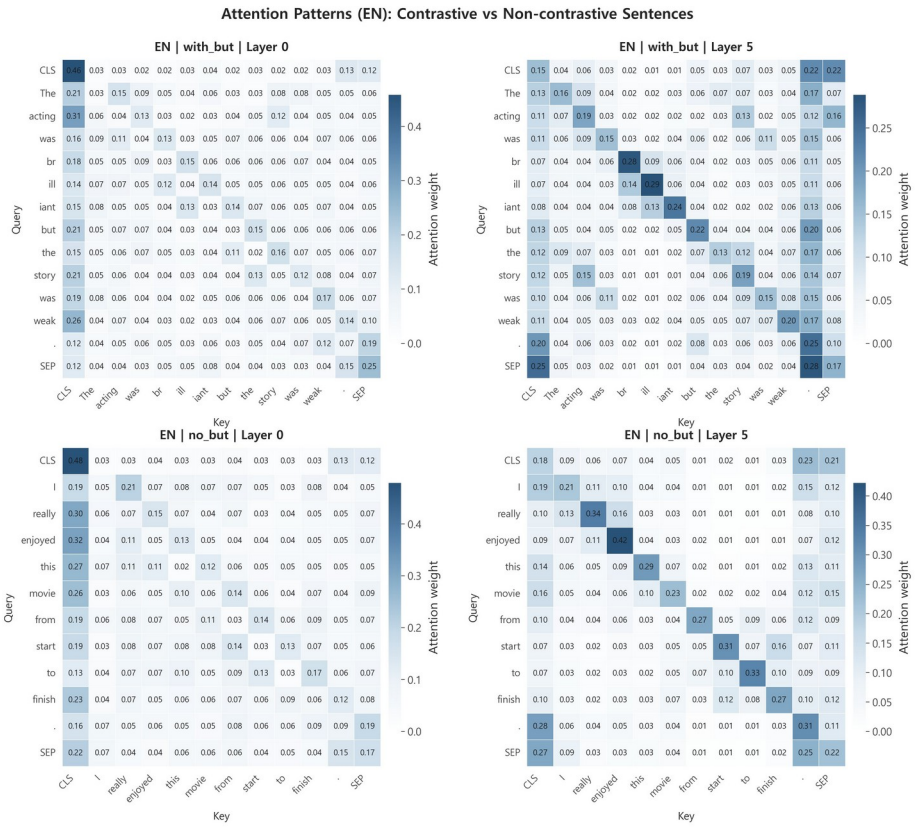


**Figure 5. English attention heatmaps at Layer 0 and Layer 5, contrastive (top, with 'but') vs sentiment (bottom).**

The Chinese pair (Figure 6) shows a similar story but with an important tokenization wrinkle. The contrastive marker 但是 is split by BPE into two tokens 但 and 是, and the sentiment marker 精彩 is split into 精 and 彩. At Layer 5 each of these character tokens attends primarily to its neighbour on the diagonal, and the multi-character phrase does not receive special collective attention. The focus-attention values in Figure 8 confirm that 但是 gets 0.061 at Layer 5 while 精彩 gets 0.048 — a small difference. Chinese attention, in this model, is organised at the character level rather than at the word level, and word-level semantic units must be reassembled in the feed-forward layers that follow attention.

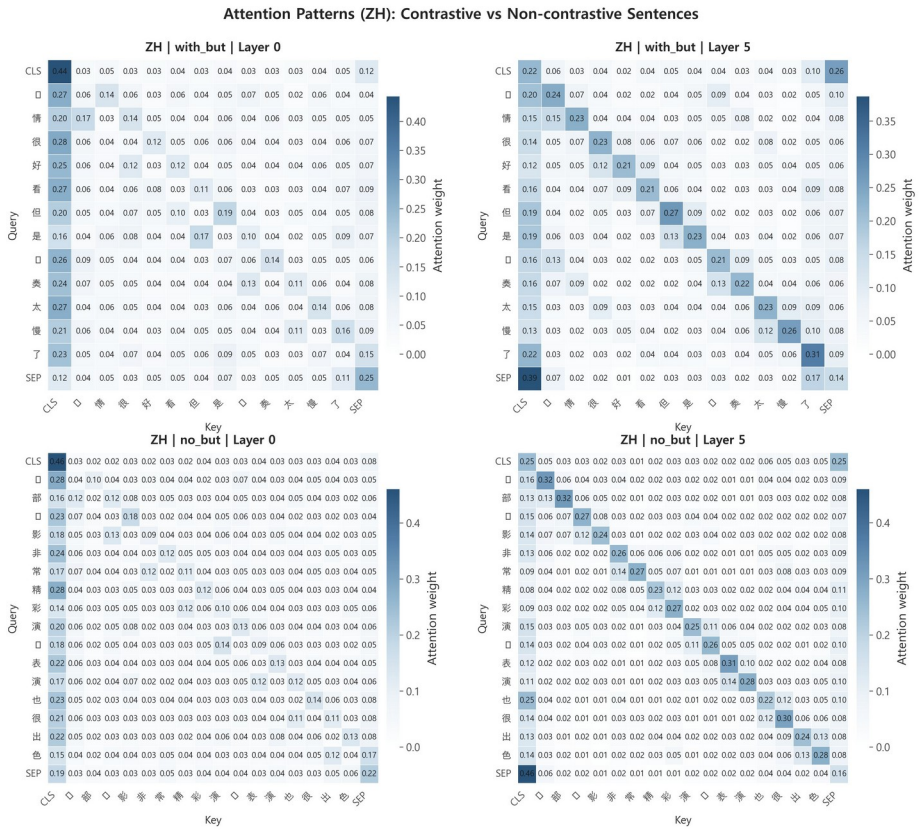


**Figure 6. Chinese attention heatmaps at Layer 0 and Layer 5, contrastive (top, with '但是') vs sentiment (bottom).**

The Korean pair (Figure 7) is the most informative of the three. The adversative suffix 지만 ("although") is tokenized as a single BPE piece, so it is directly comparable to the English but. At Layer 5 it receives 0.055 mean attention, only marginally higher than the sentiment morpheme 재미 (0.053). The attention matrix also shows a striking off-diagonal spike at 망 → SEP (0.54), i.e. the final syllable of 실망스러웠다 ("was disappointed") attends heavily to the [SEP] separator — a behavior consistent with Korean verb-final structure, where the sentence's polarity is resolved at the end of the utterance.

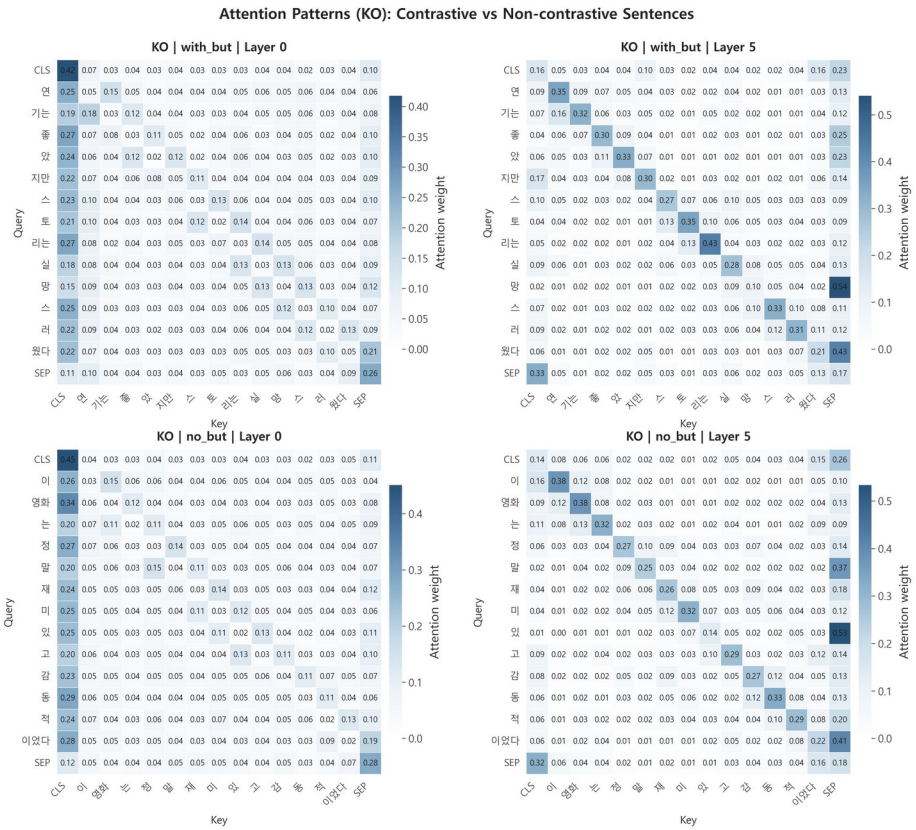


**Figure 7. Korean attention heatmaps at Layer 0 and Layer 5, contrastive (top, with '지만') vs sentiment (bottom).**

Figure 8 aggregates the numerical picture across all three languages and both layers. Several observations stand out. First, the jump from Layer 0 to Layer 5 is modest for most conditions (0.05 → 0.06 range) except for English sentiment, where enjoyed climbs from 0.067 to 0.095. Second, contrastive conjunctions do not systematically dominate attention: the Contrastive bars and the Sentiment bars are of similar height, and in English the Sentiment bar is actually higher. Third, Chinese and Korean contrastive markers receive less attention in absolute terms, almost certainly because BPE fragments them and because CJK attention is spread more uniformly across characters than English attention is across words.

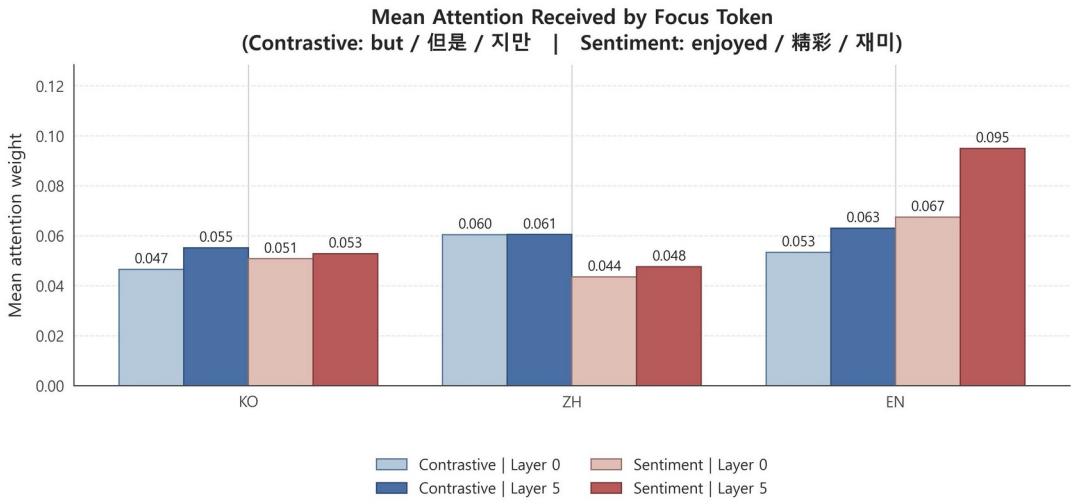


**Figure 8. Mean attention received by the focus token across languages and layers.**

Two caveats are important before any strong interpretive claim is made. First, the plotted values are averages over the twelve attention heads of DistilBERT. Individual heads are known to specialize, and it is common for one head in particular to react strongly to contrastive conjunctions while others do not; averaging washes out such specialists. Second, mean attention received is not the same as causal influence on a prediction; gradient-based measures would give a different and complementary picture. With those reservations, the most honest reading of Figures 5–8 is that the multilingual DistilBERT backbone does encode a sensitivity to contrastive structure, but that sensitivity is distributed rather than concentrated on the conjunction itself, and that sentiment content words often attract as much or more attention as the grammatical pivots around them.

# 6. Model Comparison: Base, SFT and LoRA

Three classifiers were compared on the same 80/20 stratified split (4,800 training, 1,200 test). All three share the distilbert-base-multilingual-cased backbone and a binary classifier head. They differ in which parameters are trained. The Base model freezes the backbone completely and trains only the classifier head — 592,130 parameters, or 0.44% of the 135 million total. SFT is a full fine-tune: all 135 million parameters are updated. LoRA inserts rank-8 low-rank adapters into the query and value projections of every attention layer and trains only those adapters plus the head — 739,586 trainable parameters, or 0.54%. Optimization used AdamW with learning rates of 1e-3 (Base), 2e-5 (SFT), and 3e-4 (LoRA); Base trained for six epochs, SFT for three, and LoRA for five.

The headline results are summarized in Table 3 and Figure 9.

**Table 3. Binary sentiment classification: parameters, accuracy and weighted F1 on the 1,200-document test set.**

| Model | Trainable params | % of total | Accuracy | Weighted F1 |
|---|---|---|---|---|
| Base (frozen) | 592,130 | 0.44% | 0.647 | 0.641 |

| Model | Trainable params | % of total | Accuracy | Weighted F1 |
|---|---|---|---|---|
| SFT (full FT) | 135,326,210 | 100.00% | 0.762 | 0.761 |
| LoRA (r=8, Q & V) | 739,586 | 0.54% | 0.752 | 0.752 |

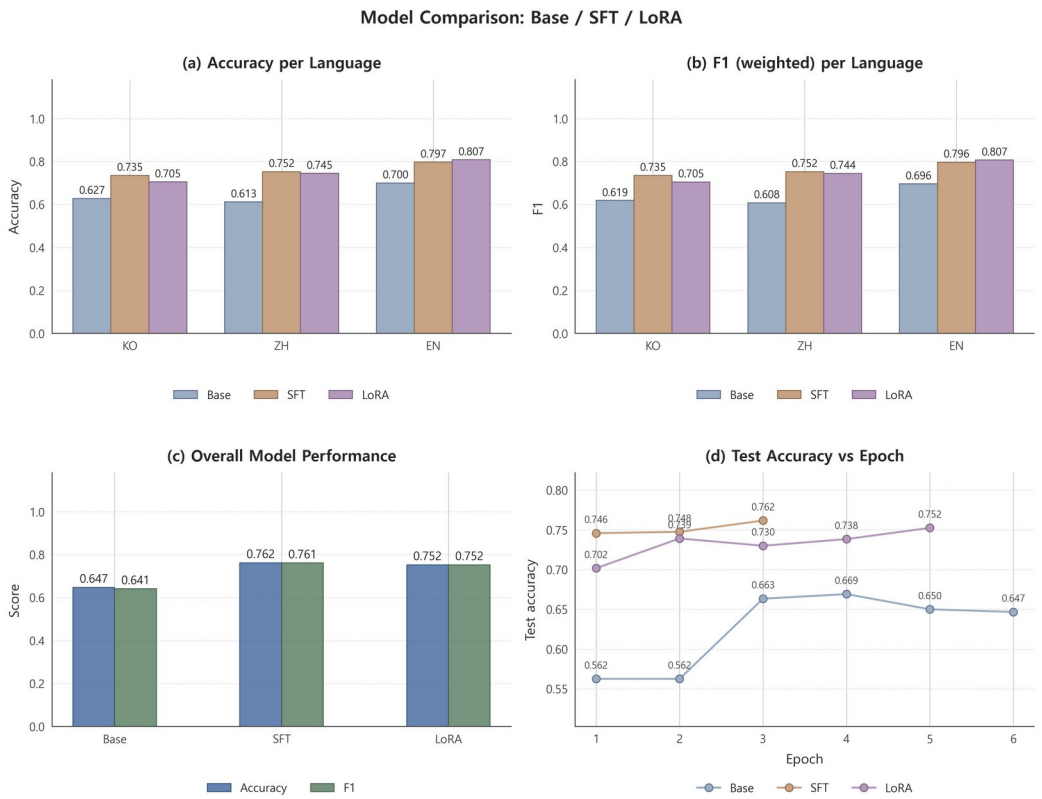


**Figure 9. Base vs SFT vs LoRA: per-language accuracy, F1, overall scores and training curves.**

The Base model reaches 64.7% accuracy, which is well above chance but clearly limited. The backbone was pretrained on general multilingual text, so its CLS embedding carries some sentiment information — enough for a linear head to pick up roughly two-thirds of the examples — but the representation was never optimized for binary polarity in reviews. This is the concrete sense in which pretraining is not enough: the model has seen vast amounts of Korean, Chinese and English text, yet its top-level features were not shaped by a sentiment objective, and so the decision boundary that a frozen backbone offers is crude.

Full fine-tuning lifts accuracy to 76.2% (+11.5 points over Base) and weighted F1 to 0.761. SFT changes every parameter in the network, and the learning curve in Figure 9(d) shows the signature of a model that is quickly specializing: the training loss drops from 0.63 to 0.28 over three epochs while test accuracy climbs from 74.6% to 76.2%. The per-language breakdown (Figure 9(a–b)) reveals that SFT benefits Chinese (+13.9 pp) and Korean (+10.8 pp) even more than English (+9.8 pp) — the base model was already most competent in English, and had more room to grow for the CJK languages.

LoRA achieves 75.2% accuracy, only 1.0 percentage point behind SFT, while updating 183 times fewer parameters. The per-language picture is more nuanced. LoRA actually beats SFT on English (80.8% vs 79.8%) but trails on Korean (70.5% vs 73.5%) and Chinese (74.5% vs 75.3%). A plausible explanation is that LoRA's low-rank restriction in Q and V acts as a regularizer: English, where the pretrained features are already close to optimal, benefits from the smaller update; the CJK languages, where the pretrained features need a larger correction, are slightly held back. That is exactly the shape of evidence one would want to produce if arguing that LoRA is not a free lunch — it is a principled bias toward small adjustments to the attention geometry.

Taken together, the three models describe the gradient between no adaptation and full adaptation. The Base model shows what the pretrained backbone offers without any task-specific learning. SFT shows what the architecture can reach when every parameter is free to move. LoRA shows that a thin low-rank perturbation of the attention mechanism captures roughly 98.7% of the SFT gain at less than one percent of the compute. For practical deployment — especially when multiple tasks must share a backbone — the LoRA result is the operationally important one.

## 7. Error Analysis

The LoRA model, being the most interesting balance between accuracy and efficiency, was selected as the subject of error analysis. It misclassified 297 out of 1,200 test examples, an overall error rate of 24.75%. Figure 10 breaks these errors down by language.

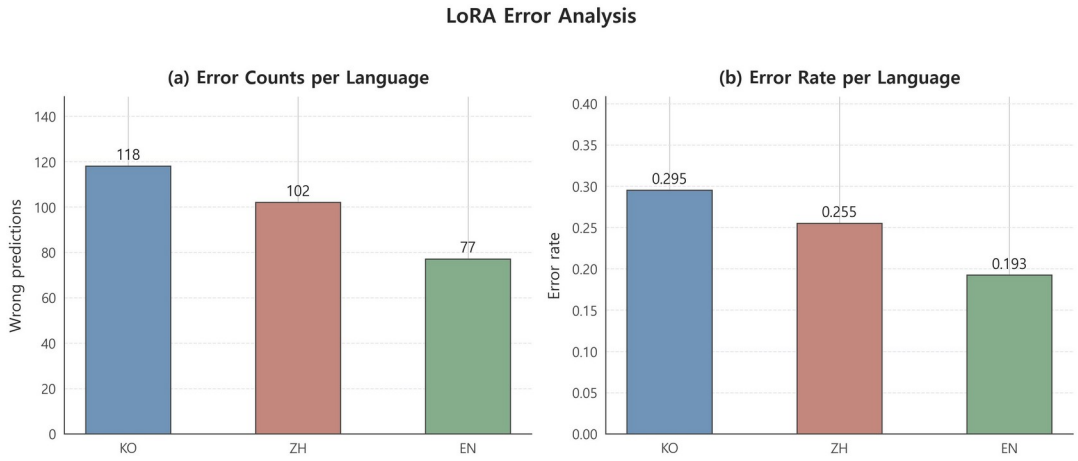


**Figure 10. LoRA error analysis: absolute counts and rates per language.**

Korean accounts for the largest share of errors (118 errors, 29.5% rate), Chinese is in the middle (102 errors, 25.5%), and English is the lowest (77 errors, 19.3%). The ranking is consistent with the tokenization and attention findings of earlier sections: Korean inflectional morphology is fragmented by the multilingual BPE and attention is spread thinly across character-level tokens; Chinese sentiment is concentrated in single characters that BPE treats individually; English enjoys both sensible whitespace tokenization and long-form context. The error rate therefore tracks how well the tokenizer preserves meaningful semantic units.

Inspecting specific misclassified examples made the pattern concrete. A Korean review labelled positive — "내용이 너무 가볍지도 않고 재밌는거같아요 보통이 마테오빠 다비드 잭히사장다맘에듬" — was predicted negative. The sentence combines a double negative ("not too light") with a fandom tail of proper names run together, and the BPE split produces a sequence of short fragments that the attention layers do not combine into a clear positive reading. A Chinese example labelled negative — a detailed complaint that the film adaptation was inferior to the novel — was predicted positive, probably because it contains many positive-valence character strings ("好", "漂亮", "睿智") that the model picked up without tracking the overall comparative frame. An English review labelled positive but written in a meandering, historically-contextualized style about a 1912 film was predicted negative because the first half reads like a preamble rather than a recommendation.

The common thread across these error types is that LoRA, like Base and SFT, is a pattern-matcher over token distributions rather than a parser of discourse. When sentiment is realized through long-range contrast, irony, or comparisons outside the sentence, the model's confidence is driven by surface token statistics and can point the wrong way. This is the usual failure mode of transformer sentiment classifiers, and it is not obviously fixable by more parameters — it likely requires either explicit contrastive pretraining or chain-of-thought reasoning at inference time.

## 8. Multilingual Insight

The individual sections of this report each addressed one stage of the pipeline. Figure 11 pulls the numbers together and normalizes each metric to its maximum across the three languages, so that shapes can be compared at a glance.

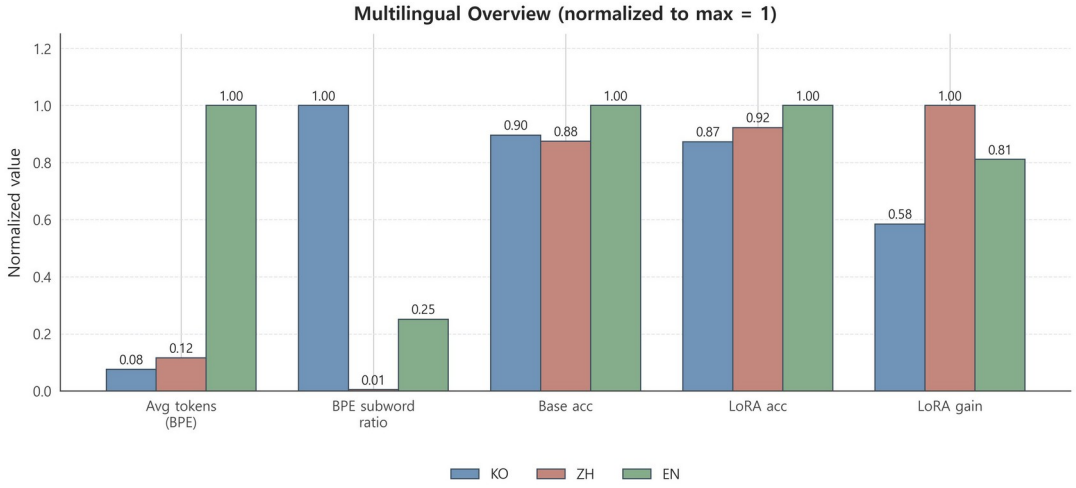


**Figure 11. Multilingual overview (normalized to each metric's maximum).**

The normalized overview makes three contrasts visible. First, the "Avg tokens (BPE)" bars show that English documents are an order of magnitude longer than Korean or Chinese documents, which matters for how much context the model receives per example. Second, the "BPE subword ratio" bars — which quantify how often multilingual BPE splits a word into two or more subwords — are nearly inverted:

Korean is at the maximum (every Hangul block is frequently decomposed), Chinese is at the minimum (characters are already vocabulary entries), and English sits in between. Third, the accuracy bars show that the Base model and the LoRA model rank languages consistently: EN > ZH > KO. The final group of bars, LoRA gain, is the most interesting. Chinese is the language that benefits most from fine-tuning (+13.3 pp from Base to LoRA), English second (+10.8 pp), Korean only third (+7.8 pp).

Reading the patterns linguistically, each language exposes a different part of the model. Korean sentiment is expressed through morphological endings glued to verb stems: 좋아요 ("I like it") vs. 좋지만 ("good, however") vs. 좋지 않아요 ("not good") are polarity-critical variations of the same root. The multilingual BPE is not optimized for Korean morphology, so these suffixes are scattered across tokens and attention is diluted. That is why Korean has the highest error rate and the smallest LoRA gain — there is an upstream bottleneck the LoRA adapters cannot fully compensate for.

Chinese sentiment is almost the opposite case. Polarity is carried by explicit content words — 好看, 精彩, 失望, 垃圾 — and tokenization is reduced to the character level. The base representation is therefore already informative per-token, and the large LoRA gain reflects the fact that a modest adaptation of attention is enough to re-weight the sentiment characters for the review-classification task. English stands between the two: polarity is expressed both lexically (amazing, terrible) and syntactically ("good but boring"), BPE preserves most words as single tokens, and attention has enough headroom to pick up the cues. The net outcome is that English has the highest ceiling in this setting, Chinese has the steepest slope of improvement, and Korean has the most resistance to a shared multilingual pipeline.

## 9. Conclusion

The study set out to understand why each stage of a multilingual NLP pipeline behaves the way it does across Korean, Chinese and English, not merely to measure how well. Five findings survive from the experiments. Tokenization dominates CJK pipelines: a shared BPE vocabulary removes the open-vocabulary problem almost completely (OOV drops from 8–12% to below 2% for every language) but in doing so fragments CJK words into characters, especially in Korean. Static Word2Vec trained on two thousand documents is insufficient to separate sentiment from co-occurrence noise, while pretrained contextual BERT embeddings adjust a word's representation by 0.20–0.40 in cosine distance depending on its context, which is exactly the property sentiment analysis needs. Attention does not automatically focus on contrastive conjunctions; content words such as enjoyed or 精彩 draw as much attention as functional pivots, and CJK attention is distributed more evenly over characters than English attention is over words. Full fine-tuning of DistilBERT reaches 76.2% accuracy on three-language sentiment; a LoRA adapter updating 0.54% of the parameters reaches 75.2%, which is 98.7% of the gain with 183× less weight movement. And finally, per-language error rates and LoRA gains correlate cleanly with tokenization quality: Chinese benefits most from adaptation because its characters are already well-tokenized; Korean benefits least because its morphology is fragmented by the shared BPE; English is bounded by a higher ceiling than either.

## 10. Current Limitations

Several boundaries of the study should be stated openly. The sample size was fixed at two thousand documents per language to keep training times comparable; larger samples would reduce the variance of the classifier and improve the quality of the Word2Vec vectors, which were clearly undertrained at this scale. Multi-movie stratification was only enforced for Chinese, because the Douban data exposed a Movie_Name_CN column; NSMC and IMDB shipped in a format that did not identify movies, so their diversity is a property of the source rather than of our sampling. The attention analysis used two sentences per language rather than a corpus-level aggregate, which keeps the visualizations readable but means that claims about attention behavior should be read as illustrative rather than statistical; a head-by-head decomposition across many sentences would be the natural extension. DistilBERT is a six-layer model, so the Layer 0 versus Layer 5 contrast is less extreme than it would be in a twelve- or twenty-four-layer BERT. Finally, the Word2Vec comparison was deliberately kept minimal — character or syllable-level embeddings might behave quite differently for Korean and Chinese — and the LoRA configuration (rank 8, applied only to query and value projections) is one point in a much larger design space that was not systematically swept.